

Bellman-Ford Algorithm

Md. Farhan Hossain
Made with KINEMASTER

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⇒ Go on relaxing with all edges.

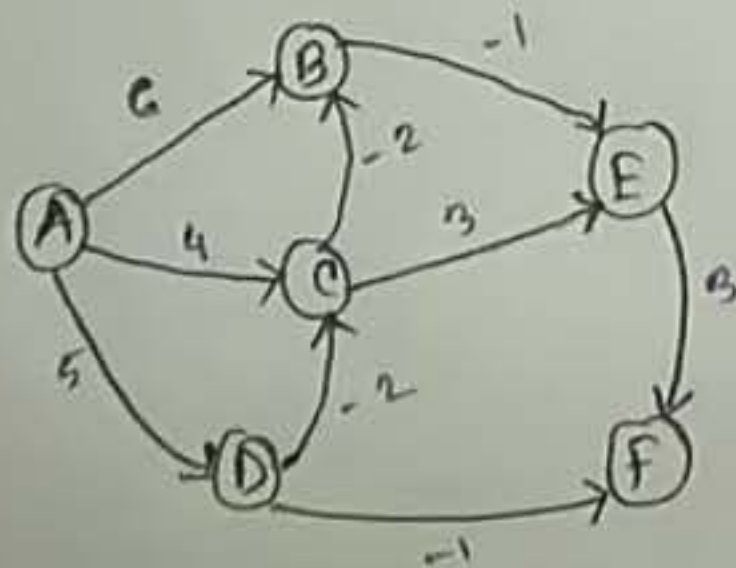
Relaxation

$$\text{If } (d(u) + c(u, v) < d(v))$$

$$d(v) = d(u) + c(u, v)$$

Iteration

$(n-1)$ [$n = \text{no of vertices}$]
 $i = 1$
 $= 5\text{-th time}$



1st

A	B	C	D	E	F

Selected vertices
↓

A	B	C

2nd

Relaxation

$$\text{if } (d(u) + c(u, v) < d(v))$$

$$d(v) = d(u) + c(u, v)$$

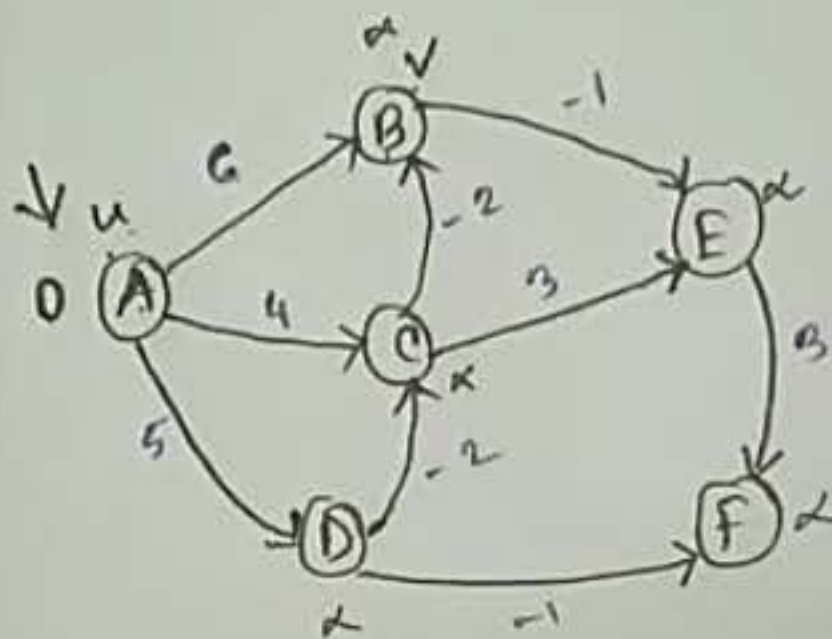
Iteration

$$(n-1) \quad [n = \text{no of vertices}]$$

$i-1$
= 5th time.

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$$\rightarrow 0 + 6 = 6 < \infty$$

Selected vertices

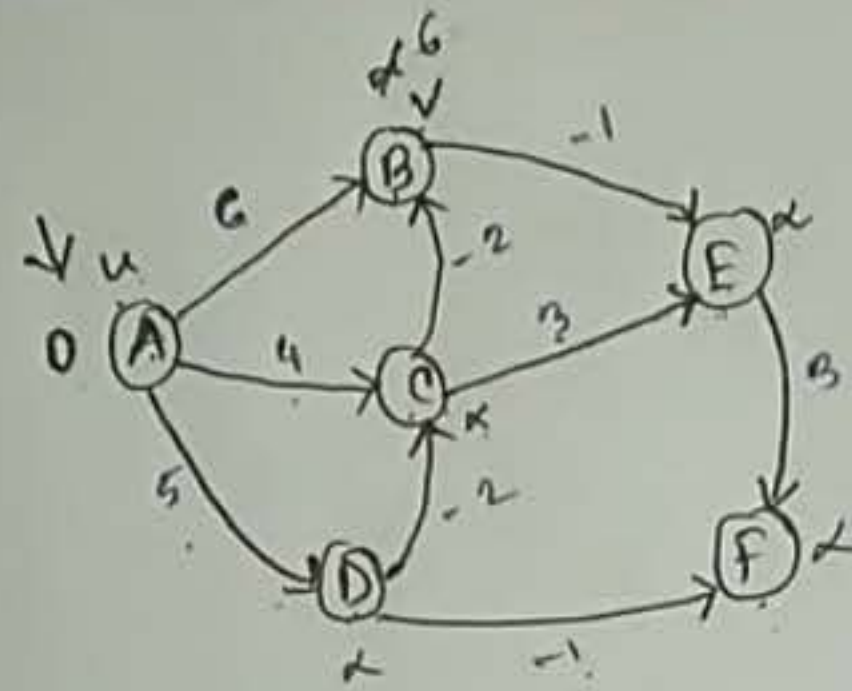
Int

	A	B	C	D	E	F
A	0	∞	∞	∞	∞	∞
B						
C						
D						
E						
F						

Selected vertices

	A	B
A		
B		
C		
D		
E		
F		

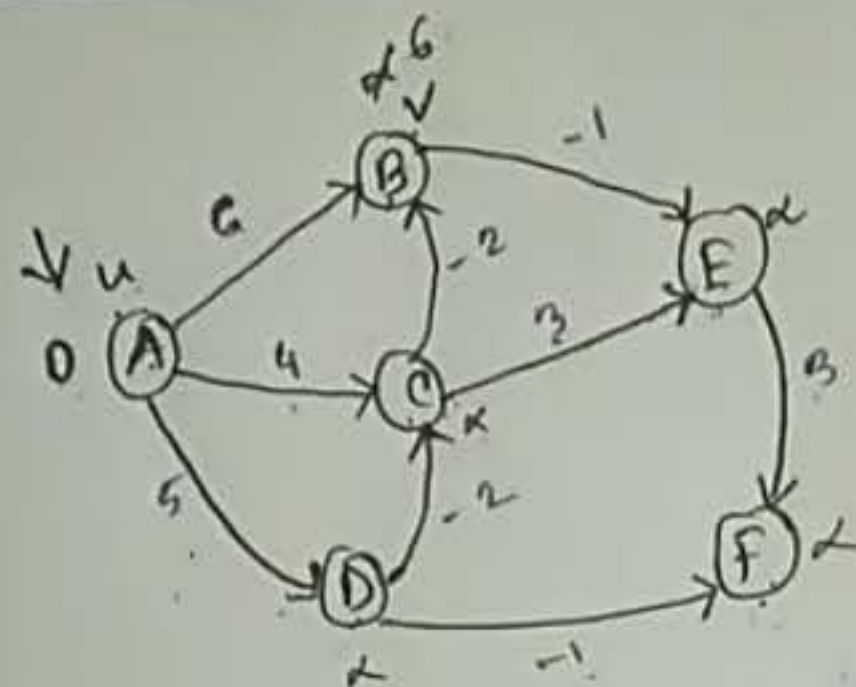
$= 5$ th time.

$$\rightarrow 0 + 6 = 6 < \infty$$
ant

	A	B	C	D	E	F
A	0	2	2	2	2	2
B	0	6	4	5	2	2
C	0	6	4	5	5	2
D	0	2	4	5	5	2
E	0	2	3	5	5	4
F	0	2	3	5	5	4

2nd

[illegible]



$$\rightarrow 0 + 6 = 6$$

$$= 6$$

Selected vertices

1st

	A	B	C	D	E	F
A	0	x	x	x	x	x
B	0	6	4	5	x	x
C	0	6	4	5	5	x
D	0	2	4	5	5	x
E	0	2	3	5	5	4
F	0	2	3	5	5	4

Selected vertices

2nd

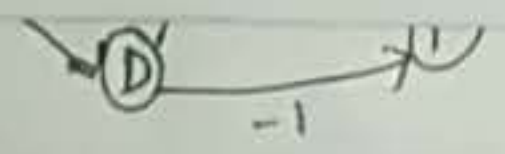
	A	B	C	D	E	F
A	0	2	3	5	5	4
B	0	2	3	5	5	4
C	0	2	3	5	1	4
D	0	1	3	5	1	4
E	0	1	3	5	1	4
F	0	1	3	5	1	4

0	<u>6</u>	4	5	∞	∞
0	6	<u>4</u>	5	5	∞
0	2	4	<u>5</u>	5	∞
0	2	3	5	<u>5</u>	4
0	<u>2</u>	3	5	<u>5</u>	<u>4</u>

A	<u>0</u>	2	3	5	5	4
B	0	<u>2</u>	3	5	5	4
C	0	2	<u>3</u>	5	1	4
D	0	1	3	<u>5</u>	1	4
E	0	1	3	5	<u>1</u>	4
F	0	1	3	5	1	<u>4</u>

Selected vertices

3rd



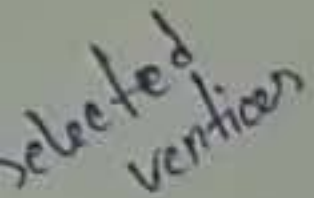
	A	B	C	D	E	F
A	<u>0</u>	1	3	5	1	4

Selected vertices

4th

	A	B	C	D	E	F

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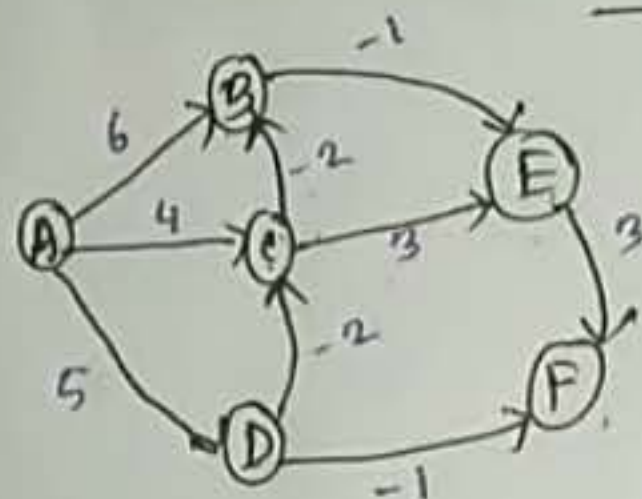


	A	B	C	D	E	F
A	0	1	3	5	1	4
B	0	1	3	5	1	4
C	0	1	3	5	0	4
D	0	1	3	5	0	4
E	0	1	3	5	0	4
F	0	1	3	5	0	3

$$4+h$$
[illegible]

Bellman-Ford Algorithm

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Selected vertices

3rd

	A	B	C	D	E	F
A	0	1	3	5	1	4
B	0	1	3	5	1	4
C	0	1	3	5	0	4
D	0	1	3	5	0	4
E	0	1	3	5	0	4
F	0	1	3	5	0	3

Selected vertices

4th

	A	B	C	D	E	F
A	0	1	3	5	0	3
B	0	1	3	5	0	3
C	0	1	3	5	0	3
D	0	1	3	5	0	3
E	0	1	3	5	0	3
F	0	1	3	5	0	3

Selected vertices

3rd



	A	B	C	D	E	F
A	<u>0</u>	1	3	5	1	4
B	0	<u>1</u>	3	5	1	4
C	0	1	<u>3</u>	5	0	4
D	0	1	3	<u>5</u>	0	4
E	0	1	3	5	<u>0</u>	4
F	0	1	3	5	0	<u>3</u>

A

We need to do 1 more iteration but since there is no change from 3rd to 4th iteration, the 5th will be the same as well. So there is no need.

Selected vertices

4th Made with KINEMASTER

	A	B	C	D	E	F
A	<u>0</u>	1	3	5	0	3
B	0	<u>1</u>	3	5	0	3
C	0	1	<u>3</u>	5	0	3
D	0	1	3	<u>5</u>	0	3
E	0	1	3	5	<u>0</u>	3
F	0	1	3	5	0	<u>3</u>

A

$$A \rightarrow 0$$

$$E \rightarrow 0$$

$$B \rightarrow 1$$

$$F \rightarrow 3$$

$$C \rightarrow 3$$

$$D \rightarrow 5$$

✓

Drawback of bellman Ford

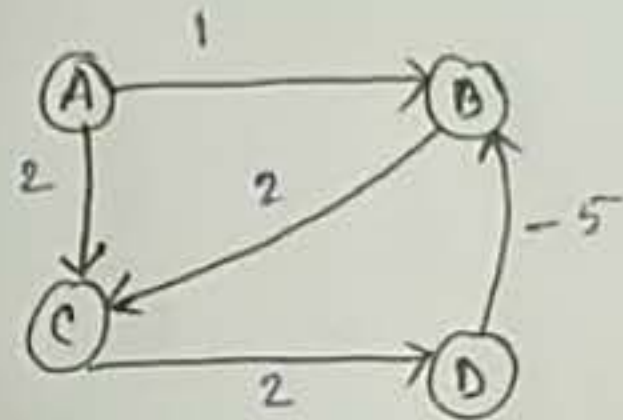
Bellman-Ford Algorithm

with **KINEMASTER**

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⇒ Drawback



Selected
vertices

ant[illegible]

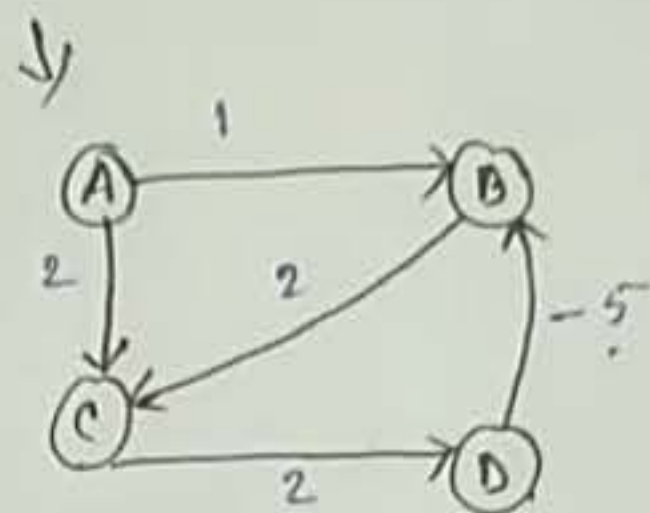
selected
vertices

2nd

A hand holding a yellow pen, pointing at a grid with columns labeled A, B, C, and D. The grid is a 4x4 table. The columns are labeled A, B, C, and D from left to right. The rows are numbered 1, 2, 3, and 4 from top to bottom. The pen is pointing at the intersection of column D and row 1.

Selected
vertices

3rd[illegible]

[illegible]

$$\begin{aligned} n-1 \\ 4-1 \\ = 3 \end{aligned}$$

Selected
vertices

ant

J	A	B	C	D
A	0	∞	∞	∞
B	0	1	2	∞
C	0	1	2	∞
D	0	1	2	4
	0	-1	2	4

selected
ventiles

2nd[illegible]

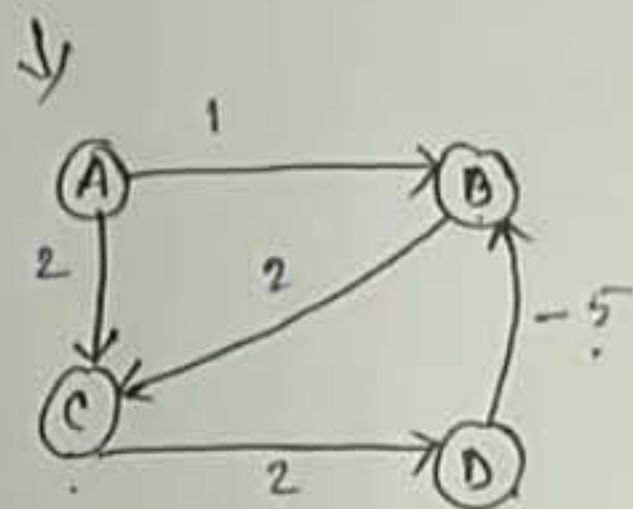
Selected
ventices

[illegible]

Selected
ventricles.

	A	B	C	D

⇒ Drawback



$$n - 1$$

$$4 - 1$$

$$= 3$$

1st

	B	C	D
A	∞	∞	∞
B	1	2	∞
C	1	2	∞
D	-1	2	4

selected
vertices

2nd

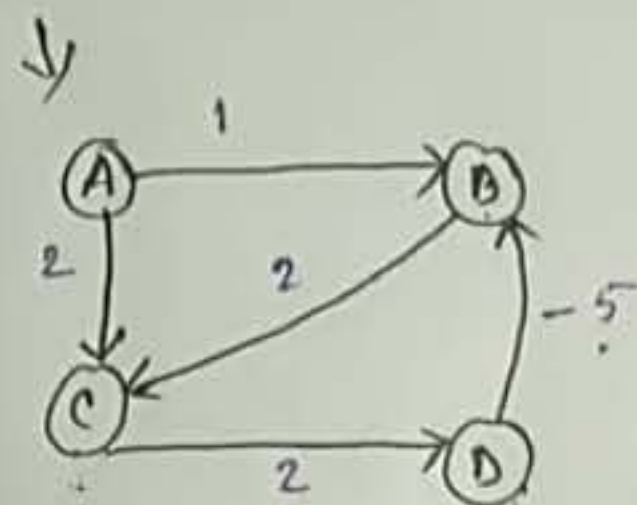
	A	B	C	D
A	0	-1	2	4
B	0	-1	2	4
C	0	-1	1	4
D	0	-1	1	3
	0	-2	1	3

selected
vertices

3rd

	A	B	C	D
A	0	-		
B				
C				
D				

⇒ Drawback



$$n - 1$$

$$4 - 1$$

$$= 3$$

1st

A	B	C	D
[0]	∞	∞	∞
0	[1]	2	∞
0	1	[2]	∞
0	1	2	[4]
0	-1	2	4

selected
vertices

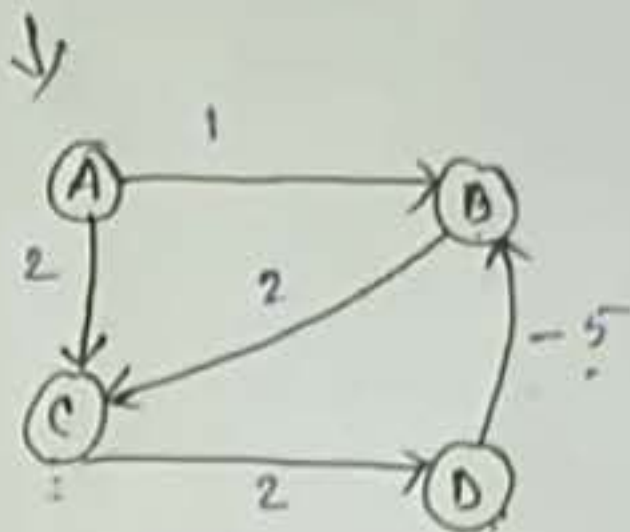
2nd

selected vertices ↓	A	B	C	D
A	[0]	-1	2	4
B	0	[-1]	2	4
C	0	-1	[1]	4
D	0	-1	1	[3]
	0	-2	1	3

selected
vertices

3rd

selected vertices ↓	A	B	C	D
A	[0]	-2	1	3
B	0	[-2]	1	3
C	0	-2	[0]	3
D	0	-2	0	[2]
	0	-3	0	2



$$4 - 1 = 3$$

Selected vertices

1st

	A	B	C	D
A	0	∞	∞	∞
B	0	1	2	∞
C	0	1	2	∞
D	0	1	2	4
	0	-1	2	4

Selected vertices

2nd

	A	B	C	D
A	0	-1	2	4
B	0	-1	2	4
C	0	-1	1	4
D	0	-1	1	3
	0	-2	1	3

Selected vertices

	A	B
A	0	-2
B	0	-2
C	0	-2
D	0	-2
	0	-3

Selected vertices

4th

	A	B	C	D
A	0	-3	0	2
B	0	-3	0	2
C	0	-3	-1	2

selected vertices

1st

	A	B	C	D
A	<u>0</u>	∞	∞	∞
B	0	<u>1</u>	2	∞
C	0	1	<u>2</u>	∞
D	0	1	2	<u>4</u>
	0	-1	2	4

selected vertices

4th

	A	B	C	D
A	<u>0</u>	-3	0	2
B	0	<u>-3</u>	0	2
C	0	-3	<u>-1</u>	2
D	0	-3	-1	<u>1</u>
	0	-4	-1	1

selected vertices

2nd

	A	B	C	D
A	<u>0</u>	-1	2	4
B	0	<u>-1</u>	2	4
C	0	-1	<u>1</u>	4
D	0	-1	1	<u>2</u>
	0	-2	1	3

selected vertices

3rd

	A	B	C	D
A	<u>0</u>	-2	1	3
B	0	<u>-2</u>	1	3
C	0	-2	<u>0</u>	3
D	0	-2	0	<u>2</u>
	0	-3	0	2

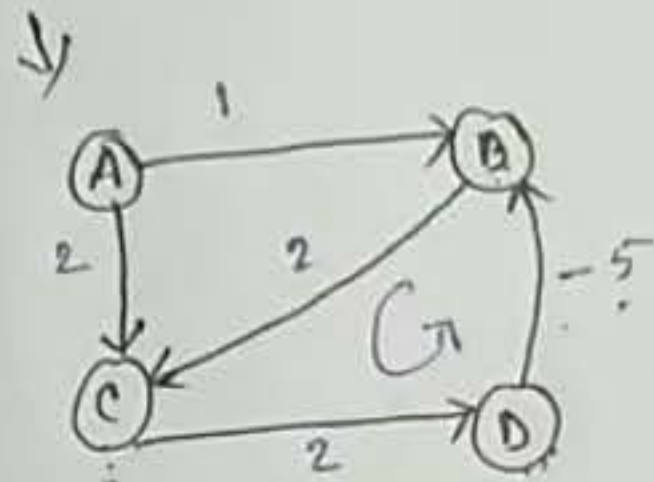
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Even after 3 iterations, unlike the rule (n-1) suggested, the 4th iteration has changes. So there is a negative weighted cycle and the Bellman Ford Algorithm will not work here.

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⇒ Drawback



$$2 + 2 - 5 = -1$$

$$4 - 1 = 3$$

Selected vertices

1st

	A	B	C	D
A	0	∞	∞	∞
B	0	1	2	∞
C	0	1	2	∞
D	0	1	2	4
	0	-1	2	4

Selected vertices

2nd

	A	B	C	D
A	0	-1	2	4
B	0	-1	2	4
C	0	-1	1	4
D	0	-1	1	3
	0	-2	1	3

Selected vertices

3rd

	A	B	C
A	0	-2	1
B	0	-2	1
C	0	-2	0
D	0	-2	0
	0	-3	0

Bellman-Ford Algorithm

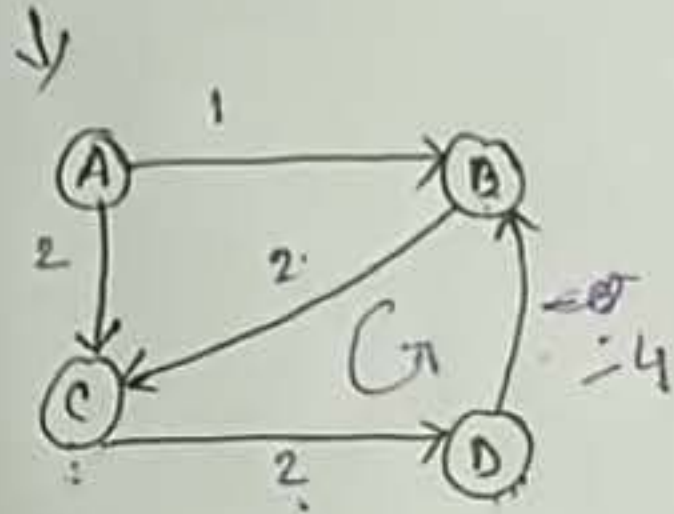
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⇒ Drawback

But is we used -4 instead of -5, there could be a valid result?



$$2 + 2 = 5$$

$$= \textcircled{-1}$$

$$n - 1$$

$$4 - 1$$

$$= \textcircled{3}$$

?

1st

	A	B	C	D
A	<u>0</u>	∞	∞	∞
B	0	<u>1</u>	2	∞
C	0	1	<u>2</u>	∞
D	0	1	2	<u>4</u>
	0	-1	2	4

selected vertices

2nd

	A	B	C	D
A	<u>0</u>	-1	2	4
B	0	<u>-1</u>	2	4
C	0	-1	<u>1</u>	4
D	0	-1	1	<u>3</u>
	0	-2	1	3

selected vertices

3rd

	A	B	C	D
A	<u>0</u>	-2	1	3
B	0	<u>-2</u>	1	3
C	0	-2	<u>0</u>	3
D	0	-2	0	<u>3</u>
	0	-3	0	3